ANNEXURE – III

VALUE ADDED COURSES

SEMESTERS I, III, IV and V

COMMON FOR ALL B.VOC PROGRAMMES

SEMESTER I

S. No.	Course Code	Course Title
1.	B.Voc. VAC 101	Introduction to Research
2.	B.Voc. VAC 102	Growing Sustainable Food in Small Spaces
3.	B.Voc. VAC 103	Nutrition, Fitness and Community Health

VALUE ADDED COURSES ADOPTED FROM OTHER FACULTIES

S. No.	Course Title
1.	Financial Literacy*
2.	Understanding Human Rights-I**

Note: *Adopted from B.COM/BBA [BCOM/BBA (VAC) – 101]
**Adopted from B.COM/BBA SEC [BCOM/BBA (SEC) – 106C]

	COURSE TITLE	INTRODUCTION TO RESEARCH
1	**Course Code**	B.Voc. VAC 101
2	**Typology of Course**	Value Added Courses (VAC)
3	**Course Duration/ LP (Lectures, Practical)**	30 hours of lectures for theory (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 Semester = 15 weeks)
4	**Credits**	2 credits
5	**Max. Marks and Time**	50 Marks (40 TH +10 IA) i. The exam of this course will be of 2 hours duration. ii. There will be no objective type questions. iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6	**Course Objectives**	To acquaint the students in understanding the relevance of research, and train them to do research in a scientific manner.
7	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Understand the importance of research methodology. CO2. Gain knowledge of different research designs. CO3. Develop an understanding of different sampling techniques. CO4. Explore date collection methods. CO5. Develop effective research writing and communication skills.
8	**Pedagogy of the Course Work**	Class Room Lecture, Interactive sessions, PPT, Group Discussions, Case Studies etc.
9	**Instructions for paper setters and candidates**	**The syllabus has been divided into TWO units**. **Note:** The question paper of each subject covering the entire course shall be divided into three sections **Section A** (10 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2.5 marks; the total weightage being 10 marks. **Section B** (15 marks) This section will consist of essay type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks. **Section C** (15 marks) This section will consist of essay type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.

10 **Course Content**

UNIT-I

Research - Meaning, Characteristics, Types, Process and Utility, Research Design - Meaning, Types and Features of good research design. Qualitative and Quantitative research. Importance of review of literature. The research proposal.

Primary and Secondary data. Methods of data collection. Measurement and scaling, Designing of Questionnaire and schedule.

UNIT-II

Sampling Design and Sampling procedure. Sampling and non-sampling errors.

Data analysis and interpretation.

Business Reports and Proposals: Introduction, Meaning, Steps in Writing a Routine Business Report, Format of Proposals, Four Key Elements in Winning Business Proposals; Case Studies.

11 **Suggested Readings**

1. Kothari, C. R. Research Methodology: Methods and Techniques. New Age International Publishers.
2. Ranjit Kumar. Research Methodology: A Step-by-Step Guide for Beginners. SAGE Publications India.
3. Panneerselvam, R. Research Methodology. PHI Learning Private Limited.
4. Creswell, J. W., & Creswell, J. D. (2017). Research Design: Qualitative, Quantitative, and Mixed Methods Approaches. SAGE Publications.
5. Neuman, W. L. Social Research Methods: Qualitative and Quantitative Approaches. Pearson.
6. Sekaran, U., & Bougie, R. Research Methods for Business: A Skill-Building Approach. Wiley.

Note: Latest edition of text books may be used.

	COURSE TITLE	GROWING SUSTAINABLE FOOD IN SMALL SPACES
1	**Course Code**	B.Voc. VAC 102
2	**Typology of Course**	Value Added Course (VAC)
3	**Course Duration/ LP (Lectures, Practical)**	15 hours of lectures for Theory + 30 hours of practicals (1 credit = 15 hours of theory learning based on 1 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4	**Credits**	2 Credits [01 Theory + 01 Practical]
5	**Max. Marks and Time**	50 Marks **Ext (Th.)** 20 | **Int (Th.)** 5 | **Ext (Pr.)** 20 | **Int (Pr.)** 5 i. The exam of this course will be of 2 hours duration. ii. There will be no objective type questions. iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
6	**Course Objectives**	To acquaint the student with the impact of food production and consumption practices on environment and economy and, to teach the student to grow some chemical free food at places of residence and work.
7	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Understand the relationship between food, health, environment and economy. CO2. Distinguish between chemical free and chemical laden food and the latter's harmful impact on health and environment. CO3. Learn how to grow their own food to some extent to be able to fulfill their nutritional needs, and to contribute their bit to save environment in the process. CO4. Learn to appreciate the symbiotic relationship between humans, plants, animals, birds and microorganisms for the survival of life forms and the Earth itself.
8	**Pedagogy of the Course Work**	Class Room Lecture, Interactive sessions, PPT, Field work with hands and Group Discussions, Case Studies etc.
9	**Instructions for paper setters and candidates**	**The syllabus has been divided into TWO units**: **Note:** The question paper of each subject covering the entire course shall be divided into three sections **Section A** (4 marks) This section will have 4 short answer questions from the entire syllabus. Students are required to attempt 2 questions from this section. Each question will carry 2 marks; the total weightage being 4 marks. **Section B** (8 marks) This section will consist of long answer type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 4 marks; the total weightage being 8 marks. **Section C** (8 marks) This section will consist of long answer type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 4 marks; the total weightage being 8 marks.

10 **Course Content**

UNIT- I

Need for Kitchen Gardens - Plants, health, environment, community, family, emotional stress busters.

Understanding types of plants as food: grains, pulses vegetables, fruits, herbs, spices, medicinal plants, oil seeds. Methods of growing plant foods: traditional, chemical dependent (green revolution in India), revert to chemical free food.

Soil –Identifying living and dead soil; Readying the soil: composting (meaning and methods)

UNIT- II

Seeds- native versus factory made. Seeds sovereignty

Seasons and food plants

Logistics - Sunlight, wind direction, season, pots, companion/adversary/support plants, and support structures like trellis.

Watering the plants and managing the nutrients of plants

Management of pests and insects

11 **Suggested Resources for Reading**

1. Fukuoka, Masanobu. The one-straw revolution: an introduction to natural farming. New York Review of Books, 2009.
2. Carson, Rachel. "Silent Spring. 1962." (2009).
3. Behera, Kambaska Kumar, et al. "Organic farming history and techniques." Agroecology and strategies for climate change (2012): 287-328.
4. Stinner, Deborah H. "The science of organic farming." (2007): 40-72.
5. Liao, Jianli, et al. "Natural farming improves soil quality and alters microbial diversity in a cabbage field in Japan." Sustainability 11.11 (2019): 3131.
6. Singh, Alka, et al. "Boosting nutritional security through kitchen gardening: Nutritional security through kitchen gardening." Journal of AgriSearch 8.3 (2021): 290-294.
7. Pavithra, G., and P. Kathirvelan. "Nutritional Kitchen Gardening." Farming at Home (2024): 77.
8. Saina, Christopher, et al. "Alleviation of Malnutrition and Food Insecurity Using Kitchen Gardening." African Journal of Education, Science and Technology 2.4 (2015): 39-48.
9. World Bank. Recipe for a Livable Planet: Achieving Net Zero Emissions in the Agrifood System, 2024. https://www.worldbank.org/en/topic/agriculture/publication/recipe-for-livable-planet

Note: Latest edition of text books may be used.

	COURSE TITLE	NUTRITION, FITNESS AND COMMUNITY HEALTH
1	**Course Code**	B.Voc. VAC 103
2	**Typology of Course**	Value Added Course (VAC)
3	**Course Duration/ LP (Lectures, Practical)**	30 hours of lectures for Theory (Theory credits – 2 (1 credit – 15 hours of theory learning based on 2hrs./week) (1 semester = 15 weeks)
4	**Credits**	2 Credits [2 TH]
5	**Max. Marks and Time**	50 Marks <table><tr><th>Ext (Th.)</th><th>Int (Th.)</th><th>Ext (Pr.)</th><th>Int (Pr.)</th></tr><tr><td>40</td><td>10</td><td>--</td><td>--</td></tr></table> i. The exam of this course will be of 2 hours duration. ii. There will be no objective type questions. iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
6	**Course Objectives**	i. To introduce the students to the fundamentals of Nutrition, food and health ii. To familiarize them with importance of nutrition during various stages of life. iii. To impart knowledge regarding etiology and management of nutritional disorders ranging from nutritional deficiencies to life style disorders.
7	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. A student after completing the course can start their own entrepreneurship. CO2. Can work in food industry. CO3. Can work in hospitals, health and fitness centers.
8	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
9	**Instructions for paper setters and candidates**	**The syllabus has been divided into TWO units:** **Note:** The question paper of each subject covering the entire course shall be divided into three sections **Section A** (8 marks) This section will have 4 short answer questions from the entire syllabus. Students are required to attempt 2 questions from this section. Each question will carry 4 marks; the total weightage being 8 marks. **Section B** (16 marks) This section will consist of long answer type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 8 marks; the total weightage being 16 marks. **Section C** (16 marks) This section will consist of long answer type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 8 marks; the total weightage being 16 marks.

10 **Course Content**

UNIT – I

Understanding Fitness

Definition of fitness, health and related terms
Assessment of fitness
Approaches for keeping fit

Importance of Nutrition

Role of nutrition in fitness
Nutritional guidelines for health and fitness
Nutritional supplements

UNIT-II

Importance of Physical Activity

Importance and benefits of physical activity
Physical Activity – frequency, intensity, time and type with examples
Physical Activity Guidelines and physical activity pyramid

Weight Management

Assessment, aetiology, health complications of overweight and obesity
Diet and exercise for weight management
Fad diets
Principles of planning weight reducing diets

11 **Suggested Resources for Reading**

1. Bamji M S, Krishnaswamy K, Brahmam G N V (2009). Textbook of Human Nutrition, 3rd edition. Oxford and IBH Publishing Co. Pvt. Ltd.
2. Srilakshmi (2010). Food Science,4th Edition. New Age International Ltd.
3. Wardlaw M G, Paul M Insel Mosby (1996). Perspectives in Nutrition, Third Edition.
4. Introduction to Human Nutrition. Ed. Gibney et al., Blackwell Publishers, 2005
5. Khanna K, Gupta S, Seth R, Mahna R, Rekhi T. (2004). The Art and Science of Cooking: A Practical Manual, Revised Edition. Elite Publishing House Pvt Ltd.

Note: Latest edition of text books may be used.

SEMESTER III

S. No.	Course Code	Course Title
1.	B.Voc. VAC 301	Business Ethics
2.	B.Voc. VAC 302	Understanding Traditional Indian Values for Administration and Management
3.	B.Voc. VAC 303	Basics of Green Chemistry

	COURSE TITLE	BUSINESS ETHICS
1	**Course Code**	B.VOC VAC 301
2	**Typology of Course**	Value Added Course (VAC)
3	**Course Duration/ LTP (Lectures, Tutorials, Practical)**	30 hours of lectures for theory (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 Semester = 15 weeks)
4	**Credits**	2 Credits Theory = 2 credits
5	**Max. Marks and Time**	50 Marks (40 TH + 10 IA) 4 × 2.5 + 4 × 7.5 = 40 Marks i. The exam of this course will be of 2 hours duration. ii. There will be no objective type questions.
6	**Course Objectives**	The objective of this paper is to familiarize the students with the importance of ethics in business and understanding of issues related to corporate social responsibility and corporate governance.
7	**Course Outcomes**	CO1. The students will be able to understand the importance of business ethics. CO2. The student will be able to grasp how to apply professional values. CO3. The students will understand the concepts of corporate social responsibility.
8	**Pedagogy of the Course Work**	Class Room Lecture, Interactive sessions, PPT, Tutorials and Group Discussions, Case Studies etc.
9	**Instructions for paper setters and candidates**	**The syllabus has been divided into TWO units.** **Note:** The question paper of each subject covering the entire course shall be divided into three sections. **Section A** (10 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2.5 marks; the total weightage being 10 marks. **Section B** (15 marks) This section will consist of essay type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks. **Section C** (15 marks) This section will consist of essay type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.
10	**Course Content**	**Unit-I** Business Ethics: Meaning and Concept, Principles of Business Ethics, Characteristics of Ethical Organisations, Theories of Business Ethics. Globalization and Business Ethics, Stakeholder's Protection, Corporate Governance and Business Ethics. Ethical Issues in Indian Business. **Unit-II** Professional Values: Objectives, Morals, Values, Ethics, Integrity, Work ethics, Service learning, Virtues, Respect for others, living peacefully, Caring, Sharing, Honesty, Courage, Valuing time,

Cooperation, Commitment, Empathy, Self-confidence, Challenges in the work place, Spirituality.

Corporate Social Responsibility: Social Responsibility of business with respect to different stakeholders, Arguments for and against social responsibility of business, Social Audit, Corporate Social Responsibility and Corporate Governance.

11 **Suggested Readings**

1. J.P. Sharma, *Corporate Governance, Business Ethics & CSR,* Ane Books Pvt. Ltd., New Delhi.
2. Andrew Crane, Dirk Matten, *Business Ethics,* Oxford University Press, New Delhi.
3. Daniel Albuquerque, *Business Ethics, Principles and Practices (Indian Edition),* Oxford University Press, New Delhi
4. Fr. Floriano C. Roa, *Business Ethics and Social Responsibility*, Rexestore.
5. O. C. Ferrell, John Fraedrich, Linda Ferrell, *Business Ethics: Ethical Decision Making & Cases,* Cengage Learning
6. Michael Blowfield, Alan Murray, *Corporate Responsibility – A Critical Introduction,* Oxford University Press, New Delhi.

Note: Latest edition of text books may be used.

	COURSE TITLE	UNDERSTANDING TRADITIONAL INDIAN VALUES FOR ADMINISTRATION AND MANAGEMENT
1	**Course Code**	B.VOC VAC 302
2	**Typology of Course**	Value Added Course (VAC)
3	**Course Duration/ LP (Lectures, Practical)**	30 hours of lectures for theory (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 Semester = 15 weeks)
4	**Credits**	2 Credits
5	**Max. Marks and Time**	50 Marks (40 TH +10 IA) i. The exam of this course will be of 2 hours duration. ii. There will be no objective type questions. iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
6	**Course Objectives**	To acquaint students with the rich indigenous value system that is relevant to workplaces – both public and private.
7	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Understand the rich tapestry of traditional Indian values and their application in administrative and managerial settings. CO2. Critically engage with ancient texts, philosophical concepts. CO3. Have practical insights to develop a deeper understanding of leadership, ethics, and sustainability from an Indian perspective. CO4. Understand challenges and opportunities in integrating traditional wisdom with modern management practices CO5. Contemplate strategies for promoting ethical leadership and sustainable management in organizations.
8	**Pedagogy of the Course Work**	Class Room Lecture, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
9	**Instructions for paper setters and candidates**	**The syllabus has been divided into TWO units**: **Note:** The question paper of each subject covering the entire course shall be divided into three sections **Section A** (10 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2.5 marks; the total weightage being 10 marks. **Section B** (15 marks) This section will consist of essay type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks. **Section C** (15 marks) This section will consist of essay type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.

10 **Course Content**

UNIT-I

Introduction to Traditional Indian Values and Philosophy

Overview of Indian philosophical traditions: Vedanta, Yoga, Sankhya, Nyaya, and Mimamsa Concepts of Dharma, Karma, and Moksha and their implications for ethical leadership

The Arthashastra: Ancient Indian Treatise on Administration and Governance

Overview of the Arthashastra by Chanakya (Kautilya)

Principles of governance, statecraft, and economic management

Application of Arthashastra principles in contemporary public administration

Leadership and Management in Indian Epics

Analysis of leadership qualities in the Ramayana and the Mahabharata

Lessons on ethical leadership, decision-making, and conflict resolution from the characters of Rama, Krishna, and Arjuna

UNIT-II

The Bhagavad Gita: Management Lessons for Life and Work

Overview of the Bhagavad Gita and its relevance to management

Concepts of duty (Dharma), detachment, and self-realization

Application of Traditional Ecological Knowledge (TEK) in maintaining modern sustainable work environment

Integration of Employee well-being and spirituality: stress management, work-life balance, mindfulness, yoga

11 **Suggested Resources for Reading**

1. Banerjee, Saikat. "Dimensions of Indian Culture Gopalan, Suresh, and Joan B. Rivera. "Gaining a perspective on Indian value orientations: Implications for expatriate managers." The International Journal of Organizational Analysis 5.2 (1997): 156-179.e, core cultural values and marketing implications: An analysis." Cross Cultural Management: An International Journal 15.4 (2008): 367-378. Chakraborty, S. K. "Man-Management in Indian Organisations: Towards a Theory of Organic Resonance." Decision 10.1 (1983): 15.
2. Chakraborty, S. K., et al. "Management paradigms beyond profit maximization." Vikalpa 29.3 (2004): 97-118.
3. Holtbrügge, Dirk, and Ritam Garg. "Indigenous Indian management philosophies: key concepts and relevance for modern Indian firms." Indian culture and work organisations in transition. Routledge India, 2016. 77-93.
4. Mukherjee, Sanjoy, and László Zsolnai, eds. Global perspectives on Indian spirituality and management: The legacy of SK Chakraborty. Springer Nature, 2022.
5. Sharma, J. K. "Relevance of ancient Indian scriptures-business wisdom drawn from Ramayana, Gita and Thirukkural." International Journal of Indian Culture and Business Management 15.3 (2017): 278-298.
6. Sharma, Subhash. "Indian ethos, Indian culture and Indian management: towards new frontiers in management thinking." Indian Culture and Management" Conference Proceedings of the ICSSR and COSMODE workshop, held at Hyderabad. 2005. Sinha, Jai BP. "Integrative indigenous management in India: Practices and prospects." Indian Journal of Industrial Relations (2000): 439-460.

Note: Latest edition of text books may be used.

	COURSE TITLE	BASICS OF GREEN CHEMISTRY
1	**Course Code**	B.VOC VAC 303
2	**Typology of Course**	Value Added Course (VAC)
3	**Course Duration/ LP (Lectures, Practical)**	30 hours of lectures for theory (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 Semester = 15 weeks)
4	**Credits**	2 Credits
5	**Max. Marks and Time**	50 Marks (40 TH + 10 IA) i. The exam of this course will be of 2 hours duration. ii. There will be no objective type questions. iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6	**Course Objectives**	To spread awareness about environmental issues and help students to use various methods of Green Synthesis
7	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Learn the importance of fossil fuels and minimize its usage. CO2. Learn and explore maximum usage of renewable sources of energy. CO3. Various methods to reduce environmental pollution for healthy life. CO4. Green methods of synthesis via cheap and naturally abundant resources which produce no hazardous waste. CO5. Solid waste management and ecofriendly methods of treatment of industrial waste.
8	**Pedagogy of the Course Work**	Class Room Lecture, Interactive sessions, PPT, Group Discussions and Case Studies etc.
9	**Instructions for paper setters and candidates**	**The syllabus has been divided into TWO units:** **Note:** The question paper of each subject covering the entire course shall be divided into three sections **Section A** (8 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2 marks; the total weightage being 8 marks. **Section B** (16 marks) This section will consist of essay type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 8 marks; the total weightage being 16 marks. **Section C** (16 marks) This section will consist of essay type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 8 marks; the total weightage being 16 marks.

10 **Course Content**

UNIT-I

Introduction: Introduction to Environment: Concept, Scope and Importance of environment. Concept of sustainability and sustainable development, Atmosphere, Lithosphere and Hydrosphere, Conservation of environment. Indian Laws concerning Environment, Biodiversity.
Renewable and Non-Renewable Resources: Land resources, Land degradation, soil erosion and desertification, Deforestation: Causes and impacts due to mining, dam building on environment, forests. Water: Use and over-exploitation of surface and groundwater, floods, droughts. Energy resources: Renewable and non-renewable energy sources, use of alternate energy sources.

UNIT-II

Environmental Pollution: Environmental pollution: types, causes, effects and controls; Air, water, soil and noise pollution. Nuclear hazards and human health risks. Solid waste management: Control measures of urban and industrial waste, Pollution case studies.

Green Chemistry for Clean Technology: Goals, Principles and application, Real cases: green synthesis of following compounds-Adipic Acid, Catechol, efficient and green synthesis of a compostable and widely applicable plastic made from corn, Healthier fats and oils by Green Chemistry.

11 **Suggested Resources for Reading**

1. Environmental Science Ceonage Learning Publications, Miller G.T. and Spool Mar, 13TH Edition, 2010.
2. Environmental Studies, Tata McGraw Hill Pub., Banny Joseph, 2018.
3. Text book of Environmental Studies for U.G. Courses-University Press–Erach Bharucna, 2nd edition, 2013.
4. Environmental Studies–From Crisis to Cure, Oxford Univ. Press, R. Raogopalan, 4th edition, 2023.
5. Environmental Chemistry (Routledge Introductions to Environment: Environmental Science), John Wright, 2003.
6. A Textbook of Green Chemistry, Publisher: Techno World, Ist edition, December 2021.
7. Green Techniques for Organic Synthesis and Medicinal Chemistry, Wei Zhang, Berkley Cue, 2nd Edition, 2018.
8. Ahluwalia, V. K. and Kidwai, M.R. New Trends in Green Chemistry, Anamalaya Publishers, 2004.
9. Advances in green synthesis, Avenues and Sustainability, Springer, 2021.

Note: Latest edition of text books may be used.

SEMESTER IV

S. No.	Course Code	Course Title
1.	B.Voc. VAC 401	Green Computing
2.	B.Voc. VAC 402	Chemistry in Everyday Life

VALUE ADDED COURSES ADOPTED FROM OTHER FACULTIES

S. No.	Course Title
1.	Right to Information and Human Rights-II*
2.	Artificial Intelligence in our Life**

Note: * Adopted from B.COM/BBA SEC Semester-II [BCOM/BBA (SEC) – 206D]

**Adopted from Journalism & Mass Communication SEC Semester-II (SCS VAC-102)

	COURSE TITLE	GREEN COMPUTING
1	**Course Code**	B.Voc VAC 401
2	**Typology of Course**	Value Added Course (VAC)
3	**Course Duration/ LP (Lectures, Practical)**	30 hours of lectures for theory (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 Semester = 15 weeks)
4	**Credits**	2 Credits
5	**Max. Marks and Time**	50 Marks (40 TH +10 IA) i. The exam of this course will be of 2 hours duration. ii. There will be no objective type questions. iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6	**Course Objectives**	This course is aimed at making students environmentally responsible by using computers and associated resources in a sustainable and responsible manner.
7	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Understand energy needs computers CO2. Learn to measure carbon footprint of computing devices. CO3. Minimize power usage by procuring sustainable hardware. CO4. Use virtualization to reduce the number of servers, and other green technologies.
8	**Pedagogy of the Course Work**	Class Room Lecture, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc. Measure computer power usage, minimize power usage, procure sustainable hardware, design green data centers, recycle computer equipment, configure computers to minimize power,
9	**Instructions for paper setters and candidates**	**The syllabus has been divided into TWO units:** **Note:** The question paper of each subject covering the entire course shall be divided into three sections **Section A** (10 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2.5 marks; the total weightage being 10 marks. **Section B** (15 marks) This section will consist of essay type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks. **Section C** (15 marks) This section will consist of essay type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.
10	**Course Content**	**UNIT-I** Introduction: Green computing fundamentals –IT in Business, Environmental Impacts of IT, Carbon Footprint, Energy Requirements of Computing devices. Life cycle of hardware - Reuse, Recycle and dispose. Green software, moving to Cloud Computing

Green Assets & Enterprises: Buildings, Data Centers, Networks, and Devices, Green Business Process Management: Modeling, Optimization, and Collaboration, Green Enterprise Architecture, Green Supply Chains, Green Information Systems: Design and Development Models.

UNIT-II

Strategies & Framework: Green Computing Strategies - IT in environmental sustainability, Drivers, Dimensions, and Goals. Green Grid Framework - Materials recycling - Green data center - Inter-organizational enterprise activities and green issues. Virtualization of IT systems- Telecommuting, teleconferencing and teleporting.

Green Compliance & Applications: Green Compliance: Policies, Protocols, Standards, Practices, and Audits – Emergent Carbon Issues: Technologies and Future. Environmentally Responsible Business Strategies (ERBS). Application & case studies of Green Computing Strategies at Home, Enterprises and Industries.

11 **Suggested Resources for Reading**

1. Bhuvan Unhelkar, Green IT Strategies and Applications-Using Environmental Intelligence‖, CRC Press, June 2014.
2. Alin Gales, Michael Schaefer, Mike Ebbers, Green Data Center: steps for the Journey, Shroff/IBM rebook, 2011.
3. Bud E. Smith, Green Computing Tools and Techniques for Saving Energy, Money, and Resources, CRC Press, June 2014.
4. Jason Harris, Green Computing and Green IT- Best Practices on regulations & industry‖, Lulu.com, 2008
5. Empowering Green Initiatives with IT‖, John Wiley & Sons, 2010.

Note: Latest edition of text books may be used.

	COURSE TITLE	CHEMISTRY IN EVERYDAY LIFE
1	**Course Code**	B.Voc VAC 402
2	**Typology of Course**	Value Added Course (VAC)
3	**Course Duration/ LP (Lectures, Practical)**	30 hours of lectures for theory (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 Semester = 15 weeks)
4	**Credits**	2 Credits
5	**Max. Marks and Time**	50 Marks (40 TH + 10 IA) i. The exam of this course will be of 2 hours duration. ii. There will be no objective type questions. iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6	**Course Objectives**	To introduce students to this fundamental branch of science that is essential to meet one's basic need of food, health, and environment. It will provide an excellent understanding of the physical universe we live in and safe use of chemical products that are used regularly.
7	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Understand the uses and applications of various compounds of chemistry. CO2. Understand importance of chemistry in various fields like food, medicine, cleansing agents etc. CO3. Understand the physical universe we live in and will be able to relate to other fields of science such as physics, biology, and environment in a better way.
8	**Pedagogy of the Course Work**	Classroom Lecture, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
9	**Instructions for paper setters and candidates**	**The syllabus has been divided into TWO units**: **Note:** The question paper of each subject covering the entire course shall be divided into three sections. **Section A** (8 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2 marks; the total weightage being 8 marks. **Section B** (16 marks) This section will consist of essay type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 8 marks; the total weightage being 16 marks. **Section C** (16 marks) This section will consist of essay type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 8 marks; the total weightage being 16 marks.

10 **Course Content**

UNIT – I

Dairy Products: Composition of milk and milk products. Analysis of fat content, minerals in milk and butter. Estimation of added water in milk.
Beverages: Analysis of caffeine in coffee and tea, estimation of methyl alcohol in alcoholic beverages.
Food Additives, Adulterants, and Contaminants: Need for additives, food preservatives like benzoates, potassium bromate, and ascorbic acid.
Artificial Sweeteners: Aspartame, Saccharin.
Flavors: Vanillin, alkyl esters (fruit flavors) and monosodium glutamate
Artificial Food Colorants: Non-permitted colours and metallic salts.
Paints and Pigments: White pigments (ZnO, TiO_2). Paints and distempers: Requirement of a good paint. Emulsion, latex; luminescent paints. Fire retardant paints and enamels, lacquers. Solvents and thinners for paints.
Dyes: Color and constitution (electronic concept). Classification of dyes. Methods of applying dyes to the fabrics. A general study of azo dyes, Congo red and methyl orange.

UNIT – II

Carbohydrates: Structure, function and Chemistry of some important mono and disaccharides.
Vitamins: Classification and Nomenclature. Sources, deficiency diseases and structures of Vitamin A, Vitamin B, Vitamin C, Vitamin D, Vitamin E and Vitamin K.
Drugs: Classification and nomenclature. Structure and function of Analgesics: Aspirin, Paracetamol. Antiallergic drug: Chlorpheniramine maleate. Antibiotics: Penicillin, Chloromycetin, Streptomycin Antimalarials: Chloroquine.
Oils and Fats: Composition of edible oils, detection of purity, rancidity of fats and oil. Tests for adulterants like mineral oils.
Soaps and Detergents: Structure and use of soaps and detergents.
Polymers: Types and classification of polymers. Source and general characteristics of natural and synthetic polymers. Typical examples of polymers used as plastics, in textiles, in electronic and automobile components, in the medical and aerospace materials. Problems of plastic waste management. Strategies for the development of environment friendly polymers.

11 **Suggested Resources for Reading**

1. B. K. Sharma: introduction to Industrial Chemistry, Goel Publishing, Meerut (2000)
2. Medicinal Chemistry by Ashtoush Kar.7th Edition, New Age Publishing, 2018
3. Drugs and Pharmaceutical Sciences Series, Marcel Dekker, Vol. II, INC, New York.
4. Analysis of Foods – H.E. Cox: 13. Chemical Analysis of Foods – H.E. Cox and Pearson.
5. Foods: Facts and Principles. N. Shakuntala Many and S. Swamy, 4th ed. New Age International Publishers.
6. Organic Chemistry by I. L. Finar, Vol. 1 & 2.
7. Polymer Science and Technology, J. R. Fired (Prentice Hall).

Note: Latest edition of textbooks may be used.

SEMESTER V

S. No.	Course Code	Course Title
1.	B.Voc. VAC 501	Total Quality Management
2.	B.Voc. VAC 502	Environmental Studies
3.	B.Voc. VAC 503	Old Age (Geriatric) Care and Counselling (OACC)
4.	B.Voc. VAC 504	Introduction to Sustainability

	COURSE TITLE	TOTAL QUALITY MANAGEMENT
1	**Course Code**	B.Voc. VAC 501
2	**Typology of Course**	Value Added Courses (VAC)
3	**Course Duration/ LP (Lectures, Practical)**	30 hours of lectures for theory (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 Semester = 15 weeks)
4	**Credits**	2 credits
5	**Max. Marks and Time**	50 Marks (40 TH + 10 IA) i. The exam of this course will be of 2 hours duration. ii. There will be no objective type questions. iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6	**Course Objectives**	To acquaint the students with the meaning and need for Total Quality Management.
7	**Course Outcomes**	On successful completion of the course, the students will be able to CO1. Understand the principles and concepts of Total Quality Management. CO2. Develop knowledge of various quality management tools and techniques used in TQM. CO3. Learn how to develop quality plans and strategies to meet customer requirements. CO4. Learn about process management and improvement.
8	**Pedagogy of the Course Work**	Class Room Lecture, Interactive sessions, PPT, Group Discussions, Case Studies etc.
9	**Instructions for paper setters and candidates**	**The syllabus has been divided into TWO units:** **Note:** The question paper of each subject covering the entire course shall be divided into three sections **Section A** (10 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2.5 marks; the total weightage being 10 marks. **Section B** (15 marks) This section will consist of essay type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks. **Section C** (15 marks) This section will consist of essay type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.

10 **Course Content**

UNIT-I

Introduction: Definition of Quality, Dimensions of Quality, Quality Planning, Quality costs and its Analysis; Basic concepts of Total Quality Management, Principles of TQM; Barriers to TQM Implementation; Quality Council and Statements; Customer Perception and Satisfaction about Quality; Customer Complaints, Service Quality; Customer Retention,
Continuous Process Improvement – Juran Trilogy, PDSA Cycle, 5S, Kaizen, Supplier Partnership –Selection and Rating

UNIT-II

Business Process Reengineering (BPR). Benchmarking –Introduction, Reasons, Process
Quality Function Deployment (QFD) – House of Quality, Benefits, Taguchi Quality Loss Function, Total Productive Maintenance (TPM).
Latest Challenges of Quality; Six Sigma Concepts: application; Quality Systems: Need for ISO 9000 and Other Quality Systems; ISO 9000:2000 Quality System
Quality Auditing; ISO 14000, Requirements and Benefits

11 **Suggested Resources for Reading**

1. Feigenbaum, A.V., Total Quality Management, McGraw-Hill Publication.
2. Oakland, J.S. & Butterworth, Total Quality Management, Hcinemann Ltd., Oxford Press.
3. Narayana, V. & Sreenivasan, N.S., Quality Management – Concepts and Tasks, New Age International.
4. Zeiri, Total Quality Management for Engineers, Wood Head Publishers.
5. James R.E., Total Quality Management, Cengage Learning.

Note: Latest edition of text books may be used.

	COURSE TITLE	ENVIRONMENTAL STUDIES
1	**Course Code**	B.Voc. VAC 502
2	**Typology of Course**	Value Added Course (VAC)
3	**Course Duration/ LP (Lectures, Practical)**	30 hours of lectures for theory (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 Semester = 15 weeks)
4	**Credits**	2 credits
5	**Max. Marks and Time**	50 Marks (40 TH + 10 IA) i. The exam of this course will be of 2 hours duration. ii. There will be no objective type questions. iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
6	**Course Objectives**	The objective of the paper is to make the student understand the basic concepts of the natural environment, to be able to identify the different sources and effects of pollution, and to have awareness about the conservation strategies and laws followed in India.
7	**Course Outcomes**	On successful completion of the course, the students will be able to CO1. Develop a comprehensive understanding of environmental issues. CO2. Demonstrate knowledge of key environmental concepts, such as biodiversity, climate change, and sustainable development. CO3. Understanding the interconnectedness of ecological systems and their role in sustaining life on Earth. CO4. Analyze the impact of human activities on the environment, including resource depletion, pollution, and habitat destruction. CO5. Develop a sense of personal and collective responsibility for environmental conservation and sustainable living practices.
8	**Pedagogy of the Course Work**	Class Room Lecture, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
9	**Instructions for paper setters and candidates**	**The syllabus has been divided into TWO units:** **Note:** The question paper of each subject covering the entire course shall be divided into three sections **Section A** (10 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2.5 marks; the total weightage being 10 marks. **Section B** (15 marks) This section will consist of essay type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks. **Section C** (15 marks) This section will consist of essay type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.

10 **Course Content**

UNIT-I

Basics of Environment: Natural resources – Meaning, Types (Air, water, soil, minerals, fossil fuels, Sunlight, humans, birds and animals, microorganisms) and Classification Systems (Abiotic vs Biotic, Exhaustible vs Non-Exhaustible, Renewable versus Non-Renewable)

Environmental systems — Meaning, categories (Climate, Lithosphere, Hydrosphere, Atmosphere, Biosphere) and their relationship

UNIT-II

Global environment problems: Ozone depletion, Climatic change, Pollution, Loss of Forests and Biodiversity and their causes

Major Environmental Laws in India : The Wildlife (Protection) Act, 1972, The Water (Prevention and Control of Pollution) Act, 1974, The Air (Prevention and Control of Pollution) Act, 1981, The Environment (Protection) Act, 1986, Biological Diversity Act 2002, Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 (FRA), The National Green Tribunal Act, 2010, Compensatory Afforestation Fund Act, 2016

11 **Suggested Resourcesfor Reading**

1. 'Ecology – Principles and Applications' by Chapman and Reiss, Cambridge University Press, Cambridge, U.K., 2004
2. 'Basic Environmental Biotechnology' by G.E. Omenn, Plenum Press, N.Y., 1987.
3. 'Environmental Challenges and Solutions' by Robert Jr. Cabin.
4. 'Global Environmental Change and Land Use' by H.J.T.M Harmaan, Jan Verhagen, C.A. Rovers, Springer, 2003.
5. 'A Textbook of Environment Science' by Thakur Vidya, Scientific Publishers, 2012.
6. 'Living in the Environment: Principles, Connections, and Solutions' by Jr., G. Tyler Miller, Brooks Cole.

Note: Latest edition of text books and authentic websites may be used.

	COURSE TITLE	OLD AGE (GERIATRIC) CARE AND COUNSELLING
1	**Course Code**	B.Voc. VAC 503
2	**Typology of Course**	Value Added Course (VAC)
3	**Course Duration/ LP (Lectures, Practical)**	30 hours of lectures for theory (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 Semester = 15 weeks)
4	**Credits**	2 Credits
5	**Max. Marks and Time**	50 Marks (40 TH + 10 IA) i. The exam of this course will be of 2 hours duration. ii. There will be no objective type questions. iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
6	**Course Objectives**	The course aims to develop an understanding of the health issues of older adults. It promotes critical thinking to gain knowledge on best practices for interventions that target health behaviours, on the individual, group, and community level. It also helps to acquire skills for health promotion among the older persons.
7	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Develop an understanding of the health issues of older adults. CO2. Understand the link between health behaviors and morbidity/ mortality. CO3. Gain knowledge about interventions that target health behaviours CO4. Actively contribute to the work of NGOs, CBOs, both national and international working in this area CO5. Initiate entrepreneurial ventures in care of elderly
8	**Pedagogy of the Course Work**	Class Room Lecture, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, Expert lectures, Online seminars, Field visits, Project reports.
9	**Instructions for paper setters and candidates**	**The syllabus has been divided into TWO units**: **Note:** The question paper of each subject covering the entire course shall be divided into three sections **Section A** (10 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2.5 marks; the total weightage being 10 marks. **Section B** (15 marks) This section will consist of essay type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks. **Section C** (15 marks) This section will consist of essay type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.

10 **Course Content**

UNIT-I

Concept of Health Promotion in Old Age
General principles of Health Education and Promotion.
Causes of morbidity and mortality in old age
Characteristic features of disease in old age.
Health Promotion – A life course approach

UNIT-II

Promotion of Physical Health in Old Age
Preventive Strategies: Vaccination, non-pharmacological and pharmacological measures; Prevention of Falls and Accidents
Lifestyle modification and Therapeutic Interventions: Fitness and Activity, Physical Exercise, Physiotherapy, Yoga, Exercise, Meditation, Weight Reduction, Calorie restriction.
Laughter club, Respite care
Hormone replacement therapy
Safer use of Medicines: Drug selection and Dosage, Drugs -Drug interaction, Drug Disease interaction.
Anti-Aging medicines

11 **Suggested Resources for Reading**

1. Sasser, J. and Moody, H. (2018). Gerontology: The basics. Routledge.
2. Quadagno, J, S. (2017). Aging and the life course: An introduction to Social Gerontology (7th Ed). McGraw Hill College.
3. Linda K. George and Kenneth F. Ferraro (2016) - Handbook of Ageing and Social Sciences, Eighth Edition. Elsevier
4. Burnard, P. (1994) Counselling Skills for Health Professionals, London: Chapman and Hall.
5. Mezey, M.D. (2001) The Encyclopedia of Elder Care, New York: Springer Publications.
6. Prakash, I.J. (2007) Strategies for an Active Old Age, Bangalore: Bangalore University.
7. Sharma, O.P. (2008) Geriatric Care – A Textbook of Geriatrics and Gerontology, New Delhi: Viva Publications.
8. WHO Regional Office for Western Pacific (2003) Aging and Health - A Health Promotion Approach for Developing Countries WHO

Note: Latest edition of text books may be used.

	COURSE TITLE	INTRODUCTION TO SUSTAINABILITY
1	**Course Code**	B.Voc. VAC 504
2	**Typology of Course**	Value Added Course (VAC)
3	**Course Duration/ LP (Lectures, Practical)**	30 hours of lectures for theory (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 Semester = 15 weeks)
4	**Credits**	2 Credits
5	**Max. Marks and Time**	50 Marks (40 TH + 10 IA) i. The exam of this course will be of 2 hours duration. ii. There will be no objective type questions. iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
6	**Course Objectives**	To make the students aware of the impact of development on the environment, socio-cultural scenario and economy, and the need for sustainable development
7	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Understand the concept of sustainability and its significance in addressing global environmental, social, and economic challenges. CO2. Evaluate the environmental impacts of human activities and identify strategies for mitigating resource depletion, pollution, and habitat destruction and promoting biodiversity. CO3. Analyze the interconnectedness of environmental, social, and economic systems and their implications for sustainable development. CO4. Examine values related to sustainability such as social justice, equity, access to resources, and the rights of future generations. CO5. Assess the role of individuals, NGOs, businesses, governments, and civil society in promoting sustainability.
8	**Pedagogy of the Course Work**	Class Room Lecture, Interactive Sessions, PPT, Practical and Group Discussions, Case Studies, etc.
9	**Instructions for paper setters and candidates**	**The syllabus has been divided into TWO units**: **Note:** The question paper of each subject covering the entire course shall be divided into three sections **Section A** (10 marks) This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 2.5 marks; the total weightage being 10 marks. **Section B** (15 marks) This section will consist of essay type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks. **Section C** (15 marks) This section will consist of essay type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 7.5 marks; the total weightage being 15 marks.

10 **Course Content**

UNIT- I

Sustainability: Meaning; Dimensions (Environmental, Socio-Cultural and Economic) and their relationship
Sustainable Development: Concept and its Significance (How present Development Model has impacted Environment)
Sustainable Development Goals: Meaning and various Goals.

UNIT -II

Socio – Cultural Sustainabilty: Processes, Structures, Amenities, Systems and Values that support equitable citizen engagement, participation and entitlement. Creation and preservation of Art, Languages, Rituals, Monuments, and other Heritage

Economic Sustainability: Macroeconomic approach (Policies, Laws and Institutions) for green investment, capital, infrastructure, employment and skills.
Inclusive and Equitable economy with a focus on Local and Indigenous economy

Sustainability Practices in Everyday Life

11 **Suggested Resources for Reading**

1. Caradonna, Jeremy L. Sustainability: A history. Oxford University Press, 2022.
2. Dhall, Meenal. "31 Indigenous Knowledge and Sustainability."
3. Docherty, Peter, Mari Kira, and AB Rami Shani, eds. Creating sustainable work systems: Developing social sustainability. Routledge, 2008.
4. Erlewein, Shina-Nancy. "Culture, Development and Sustainability: The cultural impact of development and Culture's role in sustainability." Going Beyond: Perceptions of Sustainability in Heritage Studies No. 2 (2017): 85-97.
5. Hazra, Somnath, and Anindya Bhukta. Sustainable Development Goals. Springer International Publishing, 2020.
6. Mishra, Himani, and Prateek Maheshwari. "Achieving sustainable development goals through Fourth Industrial Revolution: An Indian perspective." Indian Journal of Commerce and Management Studies 11.2 (2020): 63-75.
7. Munier, Nolberto. Introduction to sustainability. Amsterdam, The Netherlands: Springer, 2005.
8. Patel, Bimal N., and Ranita Nagar. Sustainable development and India: Convergence of law, economics, science, and politics. Oxford University Press, 2017.
9. Rao, CH Hanumantha. Agriculture, food security, poverty, and environment: essays on post-reform India. Oxford University Press, 2005.
10. Shmelev, Stanislav E. Ecological economics: sustainability in practice. Springer Science & Business Media, 2011.
11. UNEP. Supporting Resource Efficiency. https://www.unep.org/regions/asia-and-pacific/regional-initiatives/supporting-resource-efficiency/green-economy

Note: Latest edition of text books may be used.